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ISOMERIC YIELD RATIOS FOR THE REACTION ^{142}Nd(γ, *n*)^{141m,g}Nd

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Investigation of the isomeric ratios of the yields and cross-sections of photonuclear reactions with excitation energy above the giant dipole resonance makes it possible to obtain information about the density of nuclear levels and the contribution of direct processes and the mechanism of photonuclear reactions. Data on the isomeric yield ratios for photonuclear reactions are necessary for γ-activation analysis [1] – without such data it is impossible to assess the fundamental possibilities of the method and to develop an optimal procedure.

In the present work, the method of induced activity is used to investigate the dependence of the isomeric yield ratio $d(E_{\gamma\max}) = Y_m/Y_g$ of the photonuclear reaction ^{142}Nd(γ, *n*)^{141m,g}Nd on the maximum bremsstrahlung energy $E_{\gamma\max}$ in the range 14–35 MeV with step 1 MeV.

The targets were irradiated for 15–120 min with the bremsstrahlung beam from the high-current SB-50 betatron at the Research Institute of Applied Physics. The target consisted of neodymium in a natural mixture of isotopes. To increase the dose rate, the irradiation was performed inside the accelerator chamber of the betatron in a special location 12 cm from a tungsten target and combined with a K5-2A pneumatic transport apparatus (the sample delivery time to the irradiation site ~4 sec) [2].

The induced activity was measured with a semiconductor gamma spectrometer with a DGDK-63V detector. The gamma spectra were recorded with an analyzer with 4096 channels. The energy resolution of the spectrometer was ~3.5 keV (^{60}Co). Energy calibration of the gamma spectrometer was performed using the OSGI standard set of sources. The measurements were performed in a standard geometry, where the detector was graduated with respect to efficiency. The gamma spectra of the targets were measured, after a pause of 0.5–50 min, in a period of 3–120 min.

The population of the isomeric and ground levels was identified according to the γ lines 756.5 keV (^{141m}Nd: $J^{\pi} = 11/2^{-}$, $T_{1/2} = 60.3$ sec) and 1126.8 keV (^{141g}Nd: $J^{\pi} = 3/2^{+}$, $T^{1/2} = 2.6$ h). The data on the decay scheme which are necessary for analyzing the measurements were taken from [3]. The isomeric yield ratios were calculated from the formula [4]

$$d = \frac{Y_m}{Y_g} = \left[\frac{\lambda_g F_m(t)}{\lambda_m F_g(t)}\left(C\frac{N_g I_m \varepsilon_m}{N_m I_g \varepsilon_g} - p\frac{\lambda_g}{\lambda_g - \lambda_m}\right) + p\frac{\lambda_m}{\lambda_g - \lambda_m}\right]^{-1},$$

where λ_m and λ_g are the decay constants of the isomeric and ground states, respectively; $F_{m,g}(t) = [1 - \exp(-\lambda_{m,g}t_0)]\times$ $\times \exp(-\lambda_{m,g}t_n)[1 - \exp(-\lambda_{m,g}t_c)]$; C is a coefficient which accounts for the counts missed by the detecting apparatus and the position of the pulses; N_m and N_g are the numbers of detected decays of the isomeric and ground states, respectively; I is the intensity of γ rays with a prescribed energy per decay; ε is the efficiency of the spectrometer; p is the branching ratio of the γ transition; and t_0, t_n, and t_c are the irradiation, pause, and measurement times, respectively.

Figure 1 displays the results obtained. The absolute error in the isomeric yield ratios is determined by the statistical counting error in the photopeak of the γ line being measured and the γ-ray detection efficiency. The measurements showed

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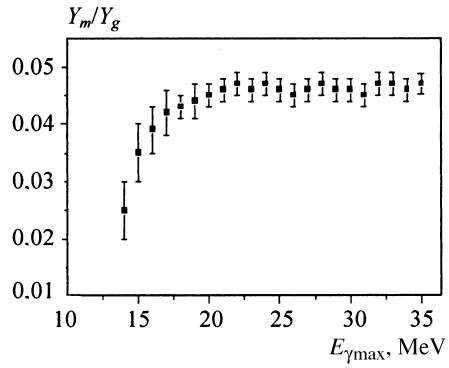


Fig. 1. Energy dependence of the isomeric yield ratio of the reaction ^{142}Nd(γ, n)^{141m,g}Nd.

TABLE 1. Isomer Yield Ratios of the Reaction ^{142}Nd(γ, n)^{141m,g}Nd

$E_{\gamma max}$, MeV	Y_m/Y_g	Source
16.5	0.045 ± 0.004	[6]
18	0.049 ± 0.004	[6]
18	0.046 ± 0.003	[7]
18	0.043 ± 0.002	This work
20	0.045 ± 0.002	Same
25	0.043 ± 0.002	[8]
30	0.045 ± 0.001	[8]
55	0.055 ± 0.0006	[9]
70	0.19 ± 0.01	[10]

that the isomeric yield ratios Y_m/Y_g increase from the reaction threshold up to the energy corresponding to the position of the maximum of the giant dipole resonance E_m, which for a photoneutron reaction on the ^{142}Nd nucleus is ~15 MeV [5]. This could be due to the fact that the number of cascade γ transitions which de-excite the nucleus increases with energy and with the angular momentum carried off by quasiprompt neutrons. For energies above E_m, the function $d(E_{\gamma max}) = Y_m/Y_g$ has the form of a saturation curve.

For energies above the giant dipole resonance, i.e., 21–35 MeV, the isomeric yield ratios of the reaction ^{142}Nd(γ, n)^{141m,g}Nd have been determined for the first time. For energies14–20 MeV, they agree, to within the limits of error, with the data of [6–8] (see Table 1). The results of [9] are somewhat higher. This is probably because in [9] they were obtained at a much higher energy (E_γ = 55 MeV). The results of [10] differ sharply from the other results. It has not been ruled out that this is due to the characteristic features of the energy dependence of isomeric ratios.

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